# HOMEWORK 2

# Histogram Processing

1. **Histogram Equalization**
   - Apply histogram equalization to Q1.jpg
   - Plot the histogram before and after equalization

Q1.jpg

2. **Histogram Specification**
   - Transform the histogram of Q2_src.jpg to match the histogram of Q2_ref.jpg
   - Plot the CDF of both images and the final mapping
   - Show the output image and its histogram

Q2_src.jpg

Q2_ref.jpg

# Spatial Filtering

## 3. Smoothing

- Apply mean filter (3×3 and 5×5) and median filter (3×3 and 5×5) to Q3.jpg
- Compare the denoising results and explain which filter works better and why

Q3.jpg

# Rules & Submission

## Rules

- Implement all algorithms using only NumPy. Do NOT use any OpenCV functions.
- You may use OpenCV only for reading/writing images (cv2.imread, cv2.imwrite).

## Submission

- Report: STUDENT_ID.pdf
  - Include Method, Result (images & comparison), Feedback
  - Max 3 pages
- Code: STUDENT_ID.zip
- Deadline: 4/7 Tue. 23:55

# Grading Policy

- Each question: 30%
- Report: 10%
- Penalties:
  - Using OpenCV functions (except reading/writing): -10 each
  - Format violation: -10
  - Report missing Method / Result img / Comparison / Feedback: -3 each
- Late submission:
  - Within 24hr: score × 0.8
  - 24hr ~ 7 days: score × 0.5